

PHYSIVERSE

Phase 6 – Agent-Based Development Architecture

Version: 1.0

Vision

Create an AI-native development system where multiple specialized agents collaborate to build, maintain, and scale Physiverse.

The system should function like a software company.

Agent Architecture

``text id="avh8fr" CEO Agent ↓ Product Manager Agent ↓ System Architect Agent ↓
Frontend Agent Backend Agent Database Agent 3D Agent AI
Agent Content Agent ————— ↓ Testing Agent ↓ Security Agent ↓
DevOps Agent ↓ Deployment Agent

Agent Principles

Every agent must:

- Be specialized.
- Own specific responsibilities.
- Never modify another agent's domain.
- Document all decisions.
- Communicate through tasks.

1. CEO Agent

Purpose:

Own the entire project vision.

Responsibilities:

- Product vision
- Business decisions
- Feature approval
- Prioritization
- Roadmap approval

Inputs:

- Product reports
- Analytics
- User feedback

Outputs:

- Product direction
- Priorities

2. Product Manager Agent

Responsibilities:

- Convert ideas into tasks
- Create epics
- Create tickets
- Define requirements
- Sprint planning

Outputs:

- Product requirements
- Task documents
- Feature specifications

3. System Architect Agent

Responsibilities:

- System design
- Folder structure
- Architecture decisions
- Tech stack decisions
- Coding standards

Outputs:

- Architecture documents
- Technical specifications

4. Frontend Agent

Responsibilities:

- UI
- Components
- Responsive layouts
- Animations
- Accessibility

Tech:

- Next.js
- Tailwind
- Framer Motion

5. 3D Experience Agent

Responsibilities:

- Three.js scenes
- React Three Fiber
- Particle systems
- Physics visualizations
- Performance optimization

Tech:

- Three.js
- React Three Fiber
- Drei
- GSAP

6. Backend Agent

Responsibilities:

- APIs
- Authentication
- Business logic
- Integrations
- Server actions

Tech:

- Next.js
- Supabase

7. Database Agent

Responsibilities:

- Schema design
- SQL migrations
- Security policies
- Optimization

Tech:

- PostgreSQL
- Supabase

8. AI Agent

Responsibilities:

- AI Tutor
- Recommendations
- Search
- Personalized learning
- Content generation

Tech:

- Gemini
- OpenAI
- Groq

9. Content Agent

Responsibilities:

- Lessons
- Simulations content
- Quizzes
- Explanations
- Documentation

10. Testing Agent

Responsibilities:

- Unit tests
- Integration tests
- E2E tests
- Accessibility tests
- Performance tests

Tech:

- Jest
- Playwright

11. Security Agent

Responsibilities:

- Authentication audits
- Authorization
- Vulnerability scanning
- Dependency analysis

12. DevOps Agent

Responsibilities:

- Docker
- CI/CD
- Monitoring
- Logging

- Infrastructure

Tech:

- Docker
- GitHub Actions
- Vercel

13. Deployment Agent

Responsibilities:

- Production deployment
- Rollbacks
- Environment management
- Monitoring

Communication System

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Task

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Architect Review

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Implementation

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Testing

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Deployment

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## Repository Structure

```text id="g9hsk1" apps/ packages/ services/ agents/ docs/ scripts/ infrastructure/

Agent Memory

Every agent maintains:

- Current tasks
- Decisions
- Dependencies
- Documentation
- Context

Task Lifecycle

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Backlog

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Planned

↓

In Progress

↓

Review

↓

Testing

↓

Done

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## Development Workflow

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Requirement

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Specification

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Architecture

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Implementation

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Testing

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Deployment

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Monitoring

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## Coding Standards

- TypeScript Strict Mode
  - ESLint
  - Prettier
  - Conventional Commits
  - Documentation Required
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## Context Management

Agents should have access to:

- Product documents
  - Architecture docs
  - Database docs
  - Previous decisions
  - Coding standards
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## Performance Targets

Lighthouse: 95+

Core Web Vitals: Excellent

First Load: <2 seconds

Accessibility: 95+

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## Final Goal

An autonomous AI software company capable of building and scaling Physiverse.